

CWA SCIENCE CLASSES

Concept With Abhishek | Class 9 — Chemistry

Chapter 1: Matter in Our Surroundings

1. Matter is anything that occupies space and has mass. It is made up of tiny particles.
2. Characteristics of particles of matter: they have space between them, they are continuously moving (kinetic energy), and they attract each other.
3. States of Matter: Solid (definite shape & volume), Liquid (definite volume, no definite shape), Gas (no definite shape or volume).
4. Interconversion of states: Melting (solid to liquid), Freezing (liquid to solid), Vaporisation (liquid to gas), Condensation (gas to liquid), Sublimation (solid to gas directly).
5. Evaporation: Surface phenomenon; occurs at all temperatures below boiling point. Factors affecting rate of evaporation: temperature, surface area, humidity, wind speed.
6. Evaporation causes cooling because particles absorb energy from surroundings to overcome force of attraction.
7. Latent Heat: The hidden heat energy required to change state of matter without change in temperature.